

Momentwork 1

$$H = W = S$$

$$\text{padding} = k // 2$$

$$\text{conv } 7 \times 7 \quad \text{stride} = 2 \xrightarrow{32} \text{maxpool } 3 \times 3 \quad S2 \quad P1 \rightarrow \frac{S}{4}$$

$$\text{conv } 5 \times 5 \quad 32 \rightarrow 64 \rightarrow \frac{S}{4}$$

$$\text{conv } 3 \times 3 \quad S2 \quad 64 \rightarrow 128 \rightarrow S/8$$

$$\text{conv } 1 \times 1 \quad 128 \rightarrow 256 \rightarrow S/8$$

$$\text{conv } 3 \times 3 \quad S2 \quad 256 \rightarrow 256 \rightarrow S/16$$

$$\text{conv } 1 \times 1 \quad 256 \rightarrow 512 \rightarrow S/16$$

Global Avg Pool, Linear $512 \rightarrow 256$
Relu

$$\text{linear } 256 \rightarrow 100$$

$$\bullet \text{ conv } 7 \times 7 \quad S2$$

$$f = \left(\frac{S}{2} \right)^2 \cdot 3 \cdot 49 \cdot 32 \cdot B \cdot 2 = \underline{2352 BS^2}$$

$$\bullet \text{ maxpool } 3 \times 3 \quad S2 \quad P1$$

$$f = \underline{16 BS^2}$$

$$\bullet \text{ conv } 5 \times 5 \quad 32 \rightarrow 64 \quad P=2$$

$$f = \left(\left\lfloor \frac{\frac{S}{4} + 2 \cdot 2 - 5}{1} \right\rfloor + 1 \right)^2 \cdot 32 \cdot 64 \cdot B \cdot 25 \cdot 2 =$$

$$= \left(\frac{S}{4} \right)^2 \cdot 256 \cdot 25 \cdot B = \underline{6400 BS^2}$$

$$\bullet \text{ conv } 3 \times 3 \quad S2 \quad 64 \rightarrow 128 \quad P=1$$

$$f = \left(\frac{S}{8} \right)^2 \cdot 64 \cdot 128 \cdot 2 \cdot 9 \cdot B = S^2 \cdot 256 \cdot 9 \cdot B = \underline{2304 BS^2}$$

$$\bullet \text{ conv } 1 \times 1 \quad 128 \rightarrow 256 \quad P=0$$

$$f = \left(\frac{S}{8} \right)^2 \cdot 128 \cdot 256 \cdot 2 \cdot B = \underline{1024 BS^2}$$

↓

- conv 3x3 s2 256 → 256 p=1

$$f = \left(\frac{S}{16}\right)^2 \cdot 256 \cdot 256 \cdot 2 \cdot 9 \cdot B = 4608 B \cdot S^2$$

- conv 1x1 256 → 512 p=0

$$f = \left(\frac{S}{16}\right)^2 \cdot 256 \cdot 512 \cdot 2 \cdot B = \frac{S^2 \cdot B \cdot 1024}{16}$$

- Global AvgPool

$$f = \left(\frac{S}{16}\right)^2 \cdot 512 \cdot B = \frac{2 \cdot B \cdot S^2}{16}$$

- linear 512 → 256

$$f = 512 \cdot 256 \cdot 2 \cdot B = 2^{18} B$$

- linear 256 → 100

$$f = 256 \cdot 100 \cdot 2 \cdot B = 51200 B$$

$$\text{Flops} = 2352 B S^2 + 16 B S^2 + 6400 B S^2 + 2304 B S^2 + 1024 B S^2 + 4608 B S^2 + \frac{S^2 B 1024}{16} + 2 B S^2 + 2^{18} B + 51200 B$$

$$\textcircled{1} \text{ Flops} = 17712 B S^2 + 313344 B$$

$$\textcircled{2} \text{ memory } (S, B) = ? \quad \text{FP32}$$

Becca

$$1) 49.32.3$$

$$6) 256 \cdot 512$$

$$2) 25 \cdot 32 \cdot 64$$

$$7) 512 \cdot 256 + 256$$

$$3) 9 \cdot 64 \cdot 128$$

$$8) 256 \cdot 100 + 100$$

$$4) 128 \cdot 256$$

$$5) 9 \cdot 256 \cdot 256$$

$$\# \text{weights} = 1040324$$

активации

$$\text{input} \quad 3 \cdot \frac{B \cdot S^2}{x}$$

$$\text{conv2} \quad 4x$$

$$\text{conv5} \quad x$$

$$\text{conv3} \quad 8x$$

$$\text{conv3} \quad 2x$$

$$\text{conv6} \quad 2x$$

$$\text{maxpool} \quad 2x$$

$$\text{conv4} \quad 4$$

$$\text{avgpool+linear} \quad C \cdot B$$

* активации $\approx 26 \cdot B \cdot S^2$

$$\textcircled{2} \text{ memory } (S, B) \approx \underbrace{26 \cdot B \cdot S^2 \cdot 4}_{\text{вход + активации}} + \overbrace{1040324 \cdot 4}^{\text{веса}} \quad (\text{байт})$$

$$\textcircled{3} \text{ latency } (S, B, \Theta) = ?$$

$$\text{latency } (S, B, \Theta) = \Theta_0 + \max \left(\text{Flops} \cdot \frac{1}{\Theta_1}, \text{Bytes} \cdot \frac{1}{\Theta_2} \right)$$

Θ_0 - запуск ядра, etc

Θ_1 - эффективн. скорость ядра Flop ГПУ

Θ_2 - скорость ~~обработки~~ чтения/записи (Byte/s)

$$\text{Bytes}(S, B) = 4 \cdot (91 \cdot B S^2 + \underbrace{2148B}_{\text{head}} + 1040324)$$

$$\textcircled{4} \text{ Energy } (S, B, \Theta)$$

$$\text{Energy } (S, B, \Theta) = \Theta_3 \cdot \text{latency } (S, B, \Theta) + \\ + \Theta_4 \cdot \text{Flops } (S, B) + \Theta_5 \cdot \text{Memory } (S, B)$$

Θ_3 - Дж/с

Θ_4 - Дж/Флоп

Θ_5 - Дж/байт